Unsupervised clustering analysis – SAS Enterprice Miner

Background

Cluster analysis is a technique that combines related data points to reveal patterns and insights without the need for labels. For instance, (Violeta & Yulia, 2025) utilized clustering to comprehend social groupings, whereas (Gupta et al., 2025) used it to group regions with comparable traffic behaviour. The k-means algorithm is a widely used method in this field. It selects a number of desired groups (k), assigns each data point to the closest group based on similarity, and then modifies the centre of each group until the clusters stabilize. (Liu, 2025) illustrated how this approach aided in the organization of extensive retail data, while (Xu, 2025) showed how it might be used to efficiently classify library text data.

The k-means algorithm is one popular clustering method. This algorithm starts by deciding how many groups (k) to divide the input into. In the context of marketing, for instance, a company may decide to divide its clientele into high, medium, and low value clusters using k=3. A similar method was shown by (Xu, 2025) in library systems, where English text items were clustered for more effective classification. The algorithm chooses initial "centroids" imaginary central points for every cluster when the number of clusters is specified. Based on mathematical distance, each data point is subsequently allocated to the nearest centroid.

Following this initial concept, the algorithm recalculates each group's centroid according to the members who were actually allocated to it. Until the cluster memberships settle and cease to fluctuate, this cycle of reassigning data points and updating centroids is repeated. (Liu, 2025) asserts that this iterative technique was very successful in arranging substantial amounts of retail transaction data into insightful clusters to facilitate market basket analysis.

Employee ID	Yearly Income ($)	Cluster (k=3)
01	10,000	High Income
02	2,500	Low Income
03	7,000	Medium Income
04	12,000	High Income
05	1,000	Low Income
06	6,000	Medium Income

Table 1: Income-Based Clustering of Employees Using K-Means (k=3)

Table 1 presents the results of a k-means clustering analysis of employee yearly income data, segmented into three distinct clusters: High Income, Medium Income, and Low Income. The algorithm grouped six employees based on similarities in their annual earnings. Employees with incomes above $10,000 (IDs 01 and 04) were categorized under the High-Income cluster. Those earning between $5,000 and $7,000 (IDs 03 and 06) formed the Medium Income group, while employees with significantly lower incomes (IDs 02 and 05) were classified as Low Income. This table demonstrates the simplicity and effectiveness of k-means clustering in identifying income-based segments within a small dataset.

Businesses, governments, and researchers can find hidden structures in data by using k-means clustering, which eliminates the need for human data analysis and allows for smarter decision-making.

Research Introduction

This project applies unsupervised learning techniques to uncover hidden structure within a breast cancer diagnostic dataset. Using clustering methods in SAS Enterprise Miner, the analysis investigates whether natural groupings emerge from cytological features measured during fine-needle aspiration biopsies. Rather than relying on predefined labels, the objective was to identify intrinsic data patterns that could support diagnostic segmentation and risk differentiation.

Research Question

Can unsupervised clustering techniques identify meaningful groupings within breast cancer cytological data that reflect underlying differences in cellular abnormality and diagnostic risk?

Research Objective

- To explore the structure of the Breast Cancer Wisconsin dataset
- To apply k-means clustering using multiple cluster specifications
- To compare manual vs automatic cluster selection methods
- To evaluate which cytological features most strongly influence segmentation
- To assess the practical interpretability of cluster outputs for diagnostic insight

What is SAS Enterprise Miner?

SAS Enterprise Miner is a visual data mining and machine learning platform used for advanced analytics. It enables structured workflows for data exploration, preprocessing, modeling, and evaluation without requiring extensive coding. In business environments, it supports scalable model experimentation, comparison, and validation within a governed analytical framework.

Methodology

1. Data Exploration

Imported dataset (699 observations, 9 diagnostic features + ID). Validated variable roles and measurement levels. Reviewed distributions and data consistency

Explore dataset

Image 1.0 Data exploration

2. Preprocessing

Logged into SAS. Created new project called A1. Created a new diagrammed called A1. From sample select file import (change Advanced Advisor to YES). Click on import file and browse to file location. Confirmed input variable configuration. Evaluated missing value handling. Ensured suitability for clustering analysis

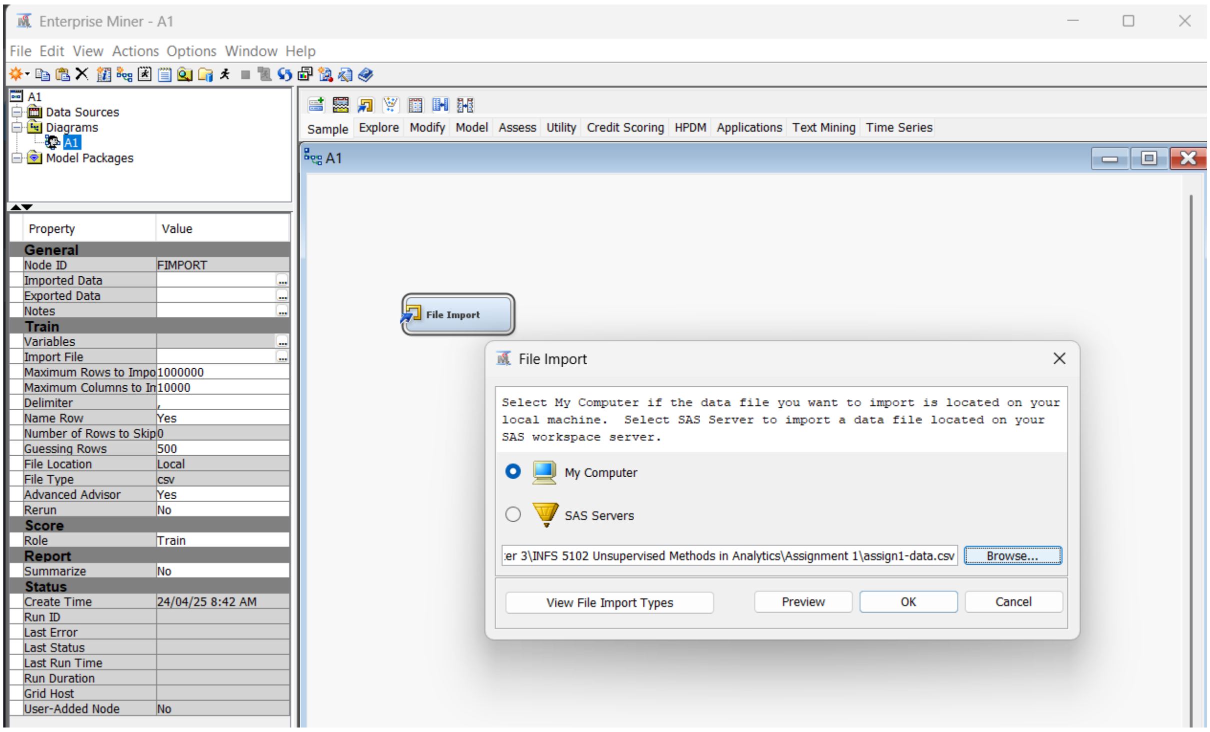


Image 2.0 file import

3. Cluster Experimentation (User-Specified k)

- Tested k = 6, 5, 4, 3, 2
- Analysed segment size, cluster profiles, and variable importance
- Observed how segmentation granularity changed as k decreased

4. Automatic Cluster Selection

- Applied Ward's Minimum Variance method
- Evaluated eigenvalue drop-off and clustering criteria
- Compared automatic selection results with manual experimentation

5. Comparative Analysis

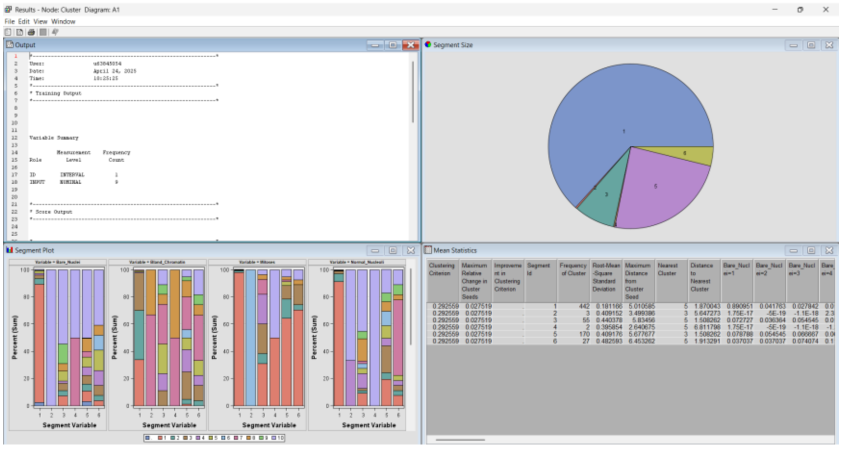
- Assessed cluster separation
- Reviewed internal variability
- Identified most influential variables across all experiments

Cluster Report K = 6

Number of Clusters

Specification Method = User Specific

Max Number of cluster = 6



Most influential:

Uniformity_Cell_Size (1.00000) and

Uniformity_Cell_Shape (0.96291)

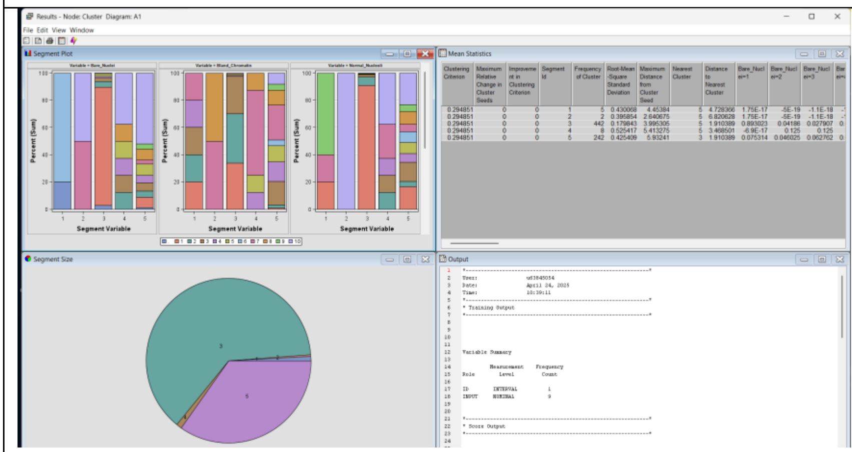
Least influential: Mitoses (0.08503)

Cluster Report K = 5

Number of Clusters

Specification Method = User Specific

Max Number of cluster = 5



Most influential variables:

Uniformity_Cell_Size (1.00000)

Uniformity_Cell_Shape (0.96915)

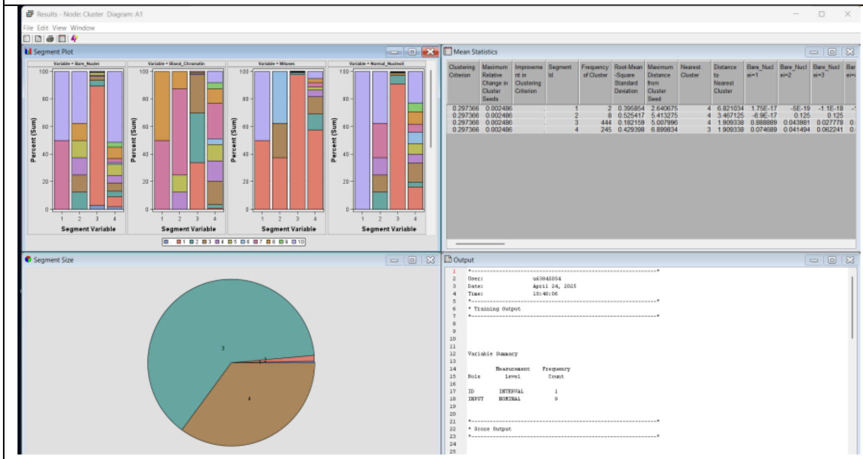
Bland_Chromatin (0.94759)

Cluster Report K = 4

Number of Clusters

Specification Method = User Specific

Max Number of cluster = 4



Most influential variables:

Uniformity_Cell_Size (1.00000)

Uniformity_Cell_Shape (0.96589)

Bland_Chromatin (0.94272)

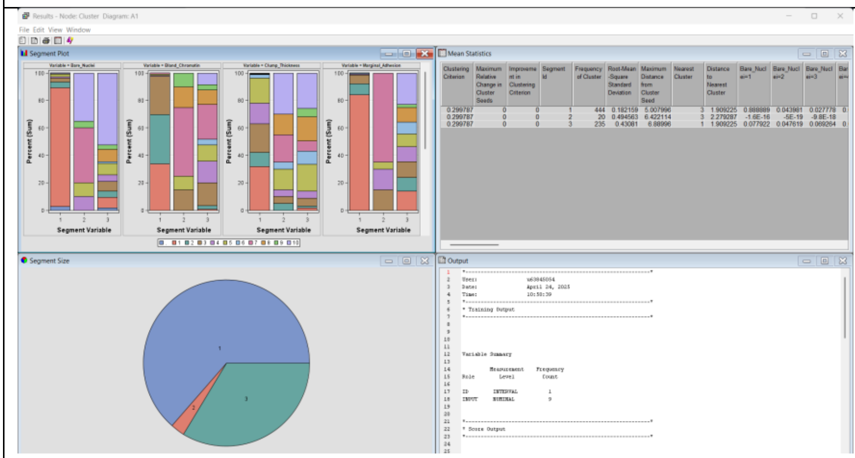
Least influential: Mitoses (0.06543)

Cluster Report K = 3

Number of Clusters

Specification Method = User Specific

Max Number of cluster = 3



Most influential variables:

Uniformity_Cell_Size (1.00000)

Uniformity_Cell_Shape (0.98679)

Bland_Chromatin (0.96273)

Moderately important: Bare_Nuclei (0.95828), Normal_Nucleoli (0.95657), S_E_Cell_Size (0.93833)

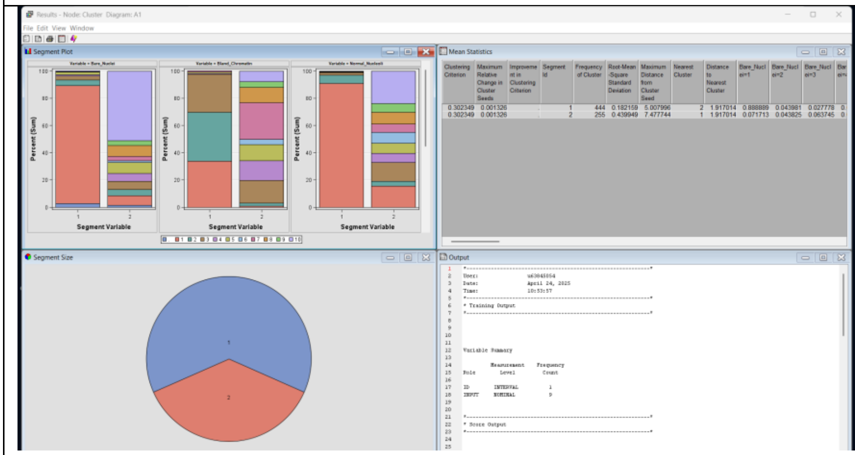
Least influential: Marginal_Adhesion (0.29497), Clump_Thickness (0.24030)

Cluster Report K = 2

Number of Clusters

Specification Method = User Specific

Max Number of cluster = 2



Most influential variables:

Uniformity_Cell_Size (1.00000)

Uniformity_Cell_Shape (0.98533)

Bland_Chromatin (0.96157)

Moderately important: Normal_Nucleoli (0.95752), S_E_Cell_Size (0.93895), Bare_Nuclei (0.93507)

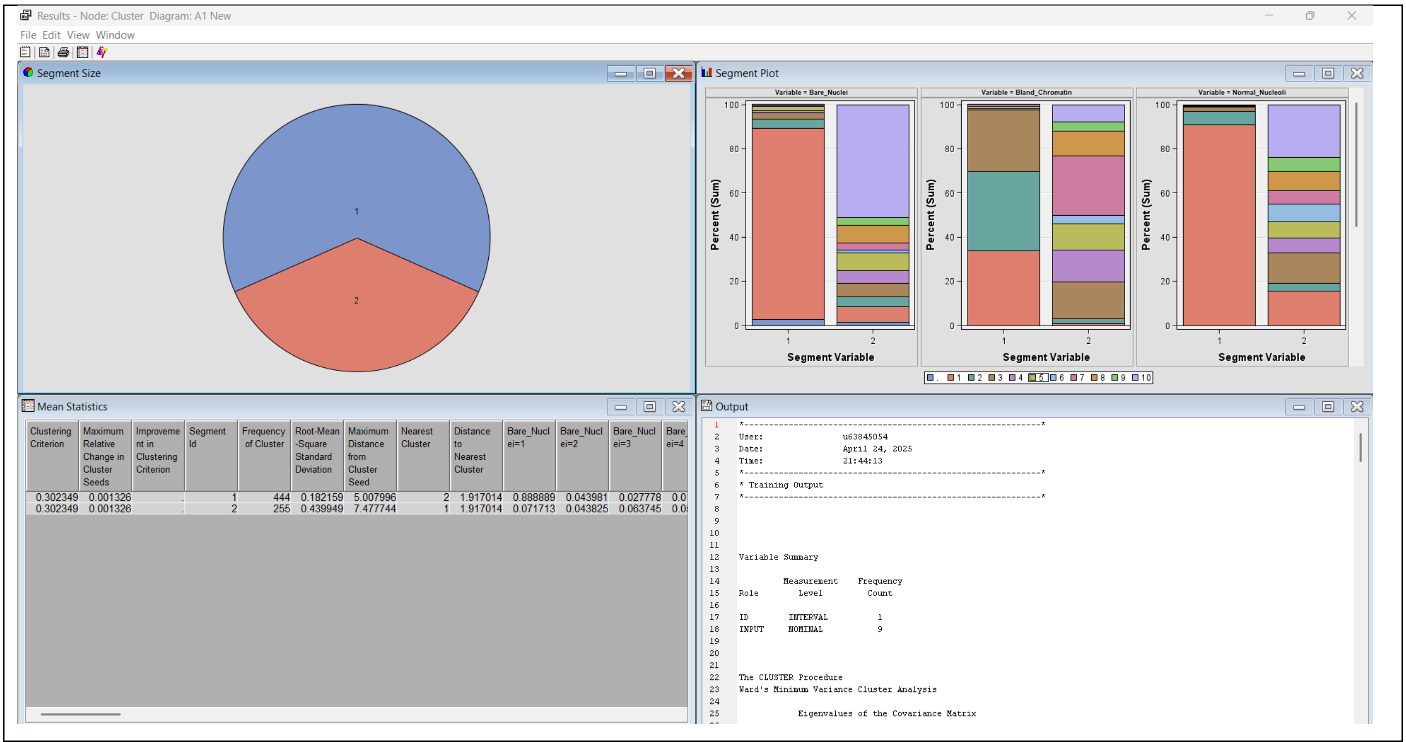
Cluster Report K = 4 (Automatic)

Finally allow SAS EM to select best number of clusters

Number of Clusters

Specification Method = Automatic

Max Number of cluster = 4 (since 4 is the average of 6-2, just a random option though)



The clustering output from SAS Enterprise Miner revealed that two clusters were automatically selected as optimal using Ward's Minimum Variance method. The eigenvalue table showed a significant drop between the first and second components (from 1.254 to 0.187), indicating that most of the variance was captured by the first principal component. This suggested that two clusters adequately explained the data structure, with minimal improvement beyond that point.

The "Segment Size" pie chart confirmed the presence of two segments, with Cluster 1 containing 444 records and Cluster 2 containing 255. Both clusters shared identical clustering criteria values (0.302349), with negligible improvement in clustering criterion change, further supporting the selection of two clusters.

The "Segment Plot" illustrated variable distributions across clusters for Bare Nuclei, Bland Chromatin, and Normal Nucleoli. Cluster 1 predominantly featured lower values across these variables, while Cluster 2 displayed a wider spread, suggesting that Cluster 2 contained cases with more pronounced abnormal features.

The mean statistics table showed that Cluster 2 had higher root-mean-square standard deviation (0.439949) and maximum distance from cluster seed (7.477744), indicating greater internal variability compared to Cluster 1. The distance between the two clusters was approximately 1.917, suggesting they were well-separated.

In summary, the system automatically selected two clusters based on internal criteria, with one cluster capturing more homogeneous cases and the other reflecting greater variation in cytological features.

Result discussion

K=6

The clustering experiment with k = 6 resulted in six distinct clusters with varying sizes and distributions. The Uniformity_Cell_Size and Uniformity_Cell_Shape were identified as the most influential variables, consistently shaping the cluster formations, while Mitoses had the least impact. The segmentation revealed detailed subgroup variations, allowing for a more granular distinction among the data points. The Segment Profile node confirmed differences across clusters, showing variations in mean values for key attributes. This clustering setup provided the most refined segmentation of the dataset, distinguishing subtle differences among patient profiles.

K=5

The clustering experiment with k = 5 resulted in five distinct clusters, showing a slightly broader segmentation compared to k = 6. Uniformity_Cell_Size remained the most influential variable, followed closely by Uniformity_Cell_Shape and Bland_Chromatin, which continued to play a significant role in defining cluster separations. The Segment Profile node highlighted differences in cluster characteristics, with some clusters showing stronger variations in key attributes. The reduction in cluster count led to a more generalized grouping, merging some previously distinct clusters while maintaining meaningful segmentation.

K=4

The clustering experiment with k = 4 resulted in four distinct clusters, showing a more generalized segmentation compared to k = 5. Uniformity_Cell_Size remained the most influential variable, followed closely by Uniformity_Cell_Shape and Bland_Chromatin, which continued to shape the cluster formations. The Segment Profile node highlighted differences in cluster characteristics, with some clusters showing stronger variations in key attributes. The reduction in cluster count led to broader groupings, merging previously distinct clusters while maintaining meaningful segmentation.

K=3

The clustering experiment with k = 3 resulted in three distinct clusters, showing a broader segmentation compared to k = 4. Uniformity_Cell_Size remained the most influential variable, followed closely by Uniformity_Cell_Shape and Bland_Chromatin, which continued to shape the cluster formations. The Segment Profile node highlighted differences in cluster characteristics, with some clusters showing stronger variations in key attributes. The reduction in cluster count led to even broader groupings, merging previously distinct clusters while maintaining meaningful segmentation.

K=2

The clustering experiment with k = 2 resulted in two broad clusters, merging previously distinct groups into larger segments. Uniformity_Cell_Size remained the most influential variable, followed closely by Uniformity_Cell_Shape and Bland_Chromatin, which continued to shape the cluster formations. The Segment Profile node highlighted differences in cluster characteristics, showing a more generalized separation of data points. The reduction in cluster count led to the most simplified segmentation, grouping similar cases together while losing finer distinctions observed in higher cluster numbers.

Comparison

K=Automatic (Max set to 4)

The clustering experiment with automatic selection and a max cluster setting of 4 resulted in two optimal clusters, as determined by Ward's Minimum Variance method. The eigenvalue table revealed a sharp drop between the first and second principal components, indicating that two clusters sufficiently captured most of the variance in the dataset. The Segment Size pie chart displayed two clusters, with Cluster 1 containing 444 records and Cluster 2 containing 255 records. Both clusters showed identical clustering criteria values, reinforcing the minimal impact of further segmentation. The Segment Plot highlighted distinct distributions in key variables, particularly Bare Nuclei, Bland Chromatin, and Normal Nucleoli, where Cluster 1 exhibited lower values while Cluster 2 displayed broader variations. The Mean Statistics table confirmed that Cluster 2 had higher root-mean-square standard deviation and greater internal variability compared to Cluster 1, suggesting that it represented cases with more pronounced cytological abnormalities. The automatic selection method validated that two clusters best represented the dataset structure, distinguishing between more homogeneous and variable cases.

Discussion

The clustering and segment profiling revealed two distinct groups within the Breast Cancer Wisconsin (Original) dataset. Cluster 1 consisted of samples with lower values across key cytological attributes. Uniformity of Cell Size, Bare Nuclei, Bland Chromatin, and Normal Nucleoli indicating more homogeneous, likely benign cell structures. Cluster 2 exhibited higher variability and elevated values in these attributes, commonly associated with malignant tumors.

These distinctions align with established cytopathological indicators of breast cancer. Malignant cells typically demonstrate irregular and enlarged nuclei, uneven chromatin distribution, and abnormal mitotic activity (Deepti et al., 2021). The clustering supported this, with Cluster 2 containing more cases likely representing malignancy due to its greater internal variability and abnormal attribute values.

The practical implication is significant: unsupervised clustering can aid in preliminary diagnostic segmentation, highlighting high-risk profiles even without labelled outcomes. This assists medical professionals in prioritizing cases for further examination. The attributes most influential in clustering. Uniformity_Cell_Size, Uniformity_Cell_Shape, and Bland_Chromatin should be emphasized in diagnostic assessments due to their consistent contribution to cluster separation.

Take-Away Knowledge

Using k-means clustering in SAS Enterprise Miner for this analysis provided valuable insights into the structure of the breast cancer dataset. A major advantage was its ability to uncover natural groupings without prior labels, helping to distinguish between likely benign and malignant cases. The method effectively utilized key cytological features like Uniformity_Cell_Size and Bland_Chromatin to form meaningful clusters, which aligned well with known medical indicators of malignancy.

SAS Enterprise Miner offered a user-friendly interface, especially useful for visually comparing clustering results through segment size charts, mean statistics, and segment profiles. This facilitated quick interpretation and pattern recognition without needing deep coding expertise.

However, the k-means algorithm was sensitive to the initial number of clusters (k), requiring trial-and-error experimentation. The need to preset k limited the method's flexibility, although the automatic clustering option using Ward's method mitigated this. Also, k-means assumes spherical clusters and equal variance, which may oversimplify real-world medical data that could have overlapped or irregular distributions.

Reference

Deepti, Ray, S., Bruckstein, A. M., Chakrabarti, A., Balas, V. E., & Sharma, N. (2021). A Review on Application of Machine Learning and Deep Learning Algorithms in Head and Neck Cancer Prediction and Prognosis. In (Vol. 70, pp. 59-73). Springer. https://doi.org/10.1007/978-981-16-2934-1_4

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Liu, L. (2025). Application of K-means supported by clustered systems in big data association rule mining. *Systems and soft computing*, *7*, 200211. https://doi.org/10.1016/j.sasc.2025.200211

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